

TranDepth: Multi-Agent Language Processing for Direct Extraction and Analysis of Images, Charts, and Formulae in Documents

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Abstract. In the context of modern digital documents that often include a combination of text, images, charts, and mathematical formulas in image format, traditional processing systems face significant limitations in comprehensively recognizing and handling these diverse components. To address this challenge, we propose a novel system based on a multi agent architecture, in which each agent is responsible for a specific task—such as image analysis, formula extraction, chart recognition, or text processing—while collaboratively operating within an interactive workflow. The breakthrough of the system lies in its ability to directly process non textual elements embedded within documents without requiring manual intervention. This approach preserves the original document structure, ensures content integrity, and provides strong support for downstream analytical tasks. Thanks to the modularity and flexibility of the agent based architecture, the system can be easily extended to accommodate various formats and complex document types in real world applications. The system was independently evaluated by three experts in AI and document processing, achieving an average score of 8.62 out of 10, significantly outperforming traditional baseline methods. The experts particularly praised the novelty of the multi agent design, its comprehensive handling capabilities, and the high accuracy in recognizing non textual content. These results demonstrate that the multi agent approach not only holds theoretical value but also delivers tangible practical benefits in the field of multimedia document processing.

Keywords: Document translation with format preservation, direct image translation within the document, in depth translation, multifunctional language processing system, AI agent based document processing workflow.

1 Introduction

In today's technological era, many popular translation systems such as Google Translate, Microsoft Translator, and new platforms like OpenL, AI TranslateDocs, and DeepL have emerged, offering online translation services. However, despite utilizing advanced technologies and artificial intelligence, these systems are still incomplete.

Their services are often fragmented and do not fully meet users' needs, particularly lacking the ability to process images directly within documents. This creates a significant gap in the market, generating a demand for more comprehensive translation solutions that can support both text and images within the same system.

Feature	Tính năng	Google Translate	Microsoft Translator	OpenL	AI TranslateDocs	DeepL	Other
Text Translate	Dịch văn bản	X	X	X		X	X
Text to Speech	Văn bản sang giọng nói	X					X
Speech to Text	Giọng nói sang văn bản	X					X
Language Analysis	Phân tích ngôn ngữ						X
Image Translate	Dịch ảnh	X		X			X
Translate documents keeping formatting	Dịch tài liệu giữ định dạng	X	X	X	X	X	X
Translate documents containing images	Dịch tài liệu chứa ảnh						

Fig. 1. Feature fragmentation in the market.

The Fourth Industrial Revolution has pushed translation from manual work to computer-, Internet-, and software-based tools (Nguyễn Thị Việt Nga, 2022). Users now need an all-in-one system that translates both plain text and text embedded in images, simplifying complex documents. This study tackles that gap by uniting core translation functions and adding automatic in-image text detection and translation. It reviews prior work, outlines the proposed architecture and workflow, and tests the image-text feature for accuracy and practicality—delivering a more efficient, unified solution for diverse needs.

2 Related Works

Modern multilingual engines—Google Translate, Microsoft Translator, DeepL, OpenL, and AI TranslateDocs—now rely on Transformer-based LLMs that outperform older MT methods, especially on specialized or context-heavy material (Miguel Angel Rios Gaona et al., 2023). These platforms bundle speech-to-text, text-to-speech, text, image, and full-document translation into a single service..

A. Speech to Text Conversion

In the field of speech to text conversion, solutions like Google Translate, Whisper, Microsoft Azure Speech to Text, AWS Transcribe, Assembly AI, and Rev.ai dominate the market. The diagram below illustrates the basic structure of a speech recognition system. Below is a summary of how the system operates.

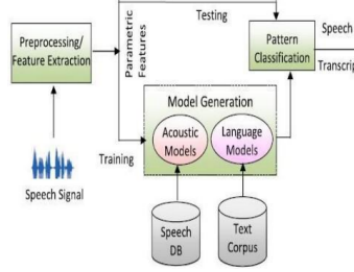


Fig. 2. Architecture for Speech Recognition System.

Source: Ayushi Trivedi et al. "Speech to text and text to speech recognition systems Areview." 2018

Speech recognition captures analog audio, digitizes it, and runs the signal through a first-order filter that equalizes the spectrum and boosts high frequencies for better accuracy. An acoustic model (phoneme-to-feature) and a language model (syntax) are co-trained on a voice database and text corpus. Together, they classify new speech samples and output the transcript (Ayushi Trivedi et al., 2018).

B. Text to Speech Conversion

Current TTS providers include Google Cloud API, Vbee, Viettel AI, and OpenAI. Through a long development history, this technology is now integrated into virtual assistants (Siri, Google Assistant, Alexa), audiobooks, public announcement systems, and video games, enhancing convenience and interaction capabilities for global users (viettelAI, 2025). In terms of the model, TTS is viewed as a function that maps text $x \rightarrow$ audio signal y , which may depend on varying parameters v (such as pitch, speed, emotion, and speaker characteristics).

$$y = f(x, v) \quad (1)$$

In which x is the input text, v is the vector of varying parameters (prosody, speaker, emotion, etc.), and y is the output sound wave. This mapping is "one to many" because the same text can produce different pronunciations depending on these variations (Nick Ovchinnikov, 2024).

C. Large Language Models (LLM)

Large language models (LLMs) such as GPT-3, PaLM, and LLaMA use Transformer stacks of multi-head attention layers, scaled to hundreds of billions of parameters and vast datasets, to achieve high-accuracy translation and other complex NLP tasks. Their performance rises sharply with increases in model size (N), data (D), and compute (C)—a relationship formalized by the KM scaling law (Kaplan et al., 2020).

$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha_N}, \quad \alpha_N \approx 0.076, \quad N_c \approx 8.8 \times 10^{13} \quad (2.1)$$

$$L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D}, \quad \alpha_D \approx 0.095, \quad D_c \approx 5.4 \times 10^{13} \quad (2.2)$$

$$L(C) = \left(\frac{C_c}{C}\right)^{\alpha_C}, \quad \alpha_C \approx 0.050, \quad C_c \approx 3.1 \times 10^8 \quad (2.3)$$

Meanwhile, a proposed Chinchilla scaling law with a loss function in the form:

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta} \quad (3)$$

with the coefficients: $E = 1.69$, $A = 406.4$, $B = 410.7$, $\alpha = 0.34$, $\beta = 0.28$. With the constraint: $C \approx 6ND$, this scaling law helps determine the optimal configuration for the model and data as follows:

$$N_{\text{opt}}(C) = G \left(\frac{C}{6} \right)^a, \quad D_{\text{opt}}(C) = G^{-1} \left(\frac{C}{6} \right)^b \quad (4)$$

In which: $a = \frac{\alpha}{\alpha+\beta}$, $b = \frac{\beta}{\alpha+\beta}$, and G is the scaling coefficient calculated from the coefficients A , B , α , and β . Chinchilla shows that best performance comes from scaling model size and data together, rather than pouring resources solely into parameters as in KM scaling (Jordan Hoffmann et al., 2022). While scaling laws predict large-model gains and guide resource use, benefits flatten near capacity, and public data may soon run out, requiring reuse or augmentation. Past ~ 60 B parameters, LLMs reveal emergent skills such as in-context learning, instruction following, and step-by-step reasoning—abilities absent in smaller models and not forecast by loss curves.

Key advances behind LLM success include precise scaling math, efficient distributed training (e.g., DeepSpeed, Megatron-LM), prompt design, instruction tuning, RLHF, and tool integration, forming a solid base for further progress (Wayne Xin Zhao et al., 2025).

The GPT series traces this evolution: GPT-1 (2018) introduced the Transformer; GPT-2 added multitask capability; GPT-3 enabled in-context learning; GPT-4 (2023) brought multimodal input and stronger safety; GPT-4 Turbo improved efficiency and added function calling. Despite their power, issues such as misinformation remain, prompting phased OpenAI deployments for safer release.

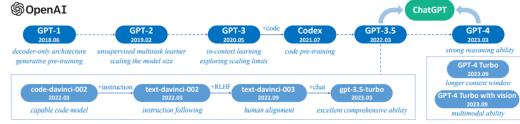


Fig. 3. A brief illustration of the technical evolution of the GPT series models.

Source: Wayne Xin Zhao et al., “A survey of large language models” *arXiv preprint arXiv:2303.18223*, 1(2).

D. OCR

Currently, systems such as Google Translate, Microsoft Translator, Online OCR, and Yandex Translate support image translation. The standard process includes: (1) OCR extracting text, (2) NMT translating the text into the target language, and (3) combining image text features through a multimodal model, attention, and shared semantic space to improve accuracy (Zhibin Lan et al., 2023). In summary, Isola's original model can be expressed as the following optimization problem.

$$\min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D) + \lambda \mathcal{L}_{L_1}(G) \quad (5.1)$$

In which $\mathcal{L}_{\text{cGAN}}$ is the loss function of a Conditional GAN (or generally, an adversarial loss), defined as follows.

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x, y \sim p_r(x, y)} [\log D(x, y)] + \mathbb{E}_{x \sim p_r(x)} [1 - \log D(x, G(x))] \quad (5.2)$$

And $\mathcal{L}_{L_1}$ is the L1 loss function (or recognition loss), defined as follows.

$$\mathcal{L}_{L_1}(G) = \mathbb{E}_{x, y \sim p_r(x, y)} [\|y - G(x)\|_1] \quad (5.3)$$

$\text{pr}(x, y)$ represents the joint distribution of paired images (e.g., Bob’s poker face vs. his genuine smile), while $\text{pr}(x)$ is the source-image distribution; G (Generator) creates synthetic images, D (Discriminator) distinguishes real from fake, and λ is the regularization factor (Xudong Pan et al., 2018). Optical Character Recognition (OCR) converts print into machine-readable text, yet modern OCR—strong on contemporary documents—often falters on older or repeatedly digitized material, producing noisy output that needs further editing (Thi Tuyet Hai Nguyen et al., 2021).

Word Selection Formula.

$$\hat{w} = \underset{w \in D}{\operatorname{argmax}} P(s|w) \times P(w) \quad (6)$$

Where:

$\hat{w}$ is the correctly chosen word.

s is the OCR token, and w is the candidate word.

$P(s | w)$ is the probability of the OCR token (s) given the word (w), known as the error model.

$P(w)$ is the probability of the word (w), known as the language model.

The goal is to find the word (w) in the dictionary (D) such that the combination of the probability of the OCR token and the probability of the word is maximized.

The Word Selection Formula helps select the correct word from the sequence of OCR tokens based on the probabilities of the words and context.

Language Model Formula.

$$P(w_1 \dots w_m) = P(w_1) \times P(w_2|w_1) \times \dots \times P(w_m|w_1 \dots w_{m-1}) \quad (7)$$

This formula describes the n gram language model, where the probability of a sequence of words $w_1, w_2, \dots, w_m$ is calculated based on the conditional probability of each word in the sequence, given the preceding words.

E. Document Translation

Document translation is booming in today’s digital age as electronic documents multiply (Giuseppe Sofo, 2024). Leading tools now include Google Translate, Microsoft Word, DeepL, AI TranslateDocs, and APIs like Google Cloud, while earlier work often used LSTM models (Ilya Sutskever, 2014).

$$p(y_1, \dots, y_{T'} | x_1, \dots, x_T) = \prod_{t=1}^{T'} p(y_t | v, y_1, \dots, y_{t-1}) \quad (8)$$

Currently, many translation tools exist, each with distinct strengths and gaps. A review of leading platforms and related software shows the field remains fragmented—highlighting the need for a single, modern system that unifies their best features and adds in-document image translation.

3 Our Proposition

This work develops a scalable AI language platform that ingests text, speech, images, and DOCX; delivers style-aware multilingual translation, accurate STT/TTS, OCR, and format preservation; and stays fast, stable, and easily extendable via plug-in mod-

els and a user-friendly UI. Its two-tier architecture separates a functional layer (user and data flow) from a logical layer where AI modules turn inputs into precise outputs.

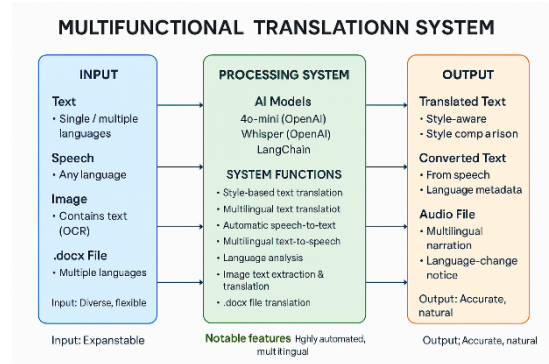


Fig. 4. The overall functional architecture diagram of the system.

The system streamlines translation and information conversion for text, speech, images, and complex documents. It handles monolingual /multilingual text (formal or informal), auto detects and transcribes speech with emotion cues, extracts text via OCR, and accurately parses .docx layouts. Using GPT 4o mini, Whisper, and LangChain, it delivers high accuracy STT, TTS, translation, semantic analysis, and OCR. Outputs range from translated text and synthesized speech to format preserved documents, giving users a reliable, versatile end to end solution.

A. Processing Flow Description

The platform's layered design maximizes speed, scale, and security. A ReactJS front end gathers text, files, and audio, forwarding them to a Node.js proxy that hides backend details, adds API keys, enforces CORS, and rate-limits traffic. A FastAPI gateway (Pydantic-validated, JWT-secured) routes tasks to GPT-4o mini for translation, Whisper for STT, EasyOCR for image text, or a 15-agent DOCX pipeline, queuing long jobs by ID. Results are logged in MySQL and returned as uniform JSON. Four core services—LangChain coordinator, Whisper, EasyOCR, and GPT-4o mini—deliver a robust, scalable, and secure workflow.

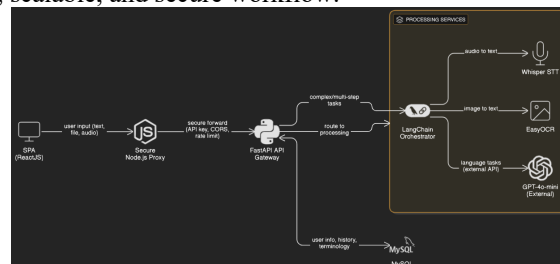


Fig. 5. The logical and technical architecture diagram.

LangChain, an open-source framework for orchestrating LLMs and tools, coordinates the system’s workflow: it ingests Whisper’s transcript, preprocesses it, calls GPT-4o mini for translation, reformats the result, and returns it to the user.

B. Speech Input Processing

The platform blends leading AI models, open-source frameworks, and specialty libraries to support in-document image processing. It begins with OpenAI Whisper for speech recognition, then adds NLP and computer-vision models to keep the system stable and scalable.

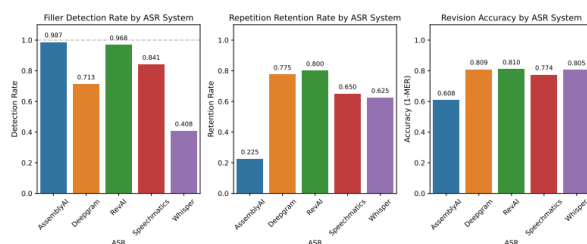


Fig. 6. Comparing the performance of speech recognition systems (ASR).

Source: Michael McGuire. “Automatic Speech Recognition for Non native English: Accuracy and Disfluency Handling.” *arXiv*, 2025.

The project uses an advanced open-source speech-to-text model that auto-detects language and remains accurate in noisy settings—ideal for real-world, multilingual audio. Within the system, it captures incoming audio and converts it to text for downstream analysis and processing.

C. Text Input Processing

OpenAI’s GPT-4o mini is the system’s language engine. This multimodal LLM combines deep contextual understanding with high-quality translation, style control, and broad NLP capabilities, all at a moderate compute cost. It handles translation, summarization, Q&A, content analysis, and on-demand text generation.

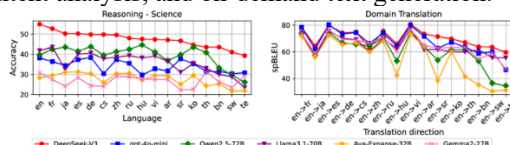


Fig. 7. Chart of language processing capabilities between large language models.

Source: Xu Huang et al. “BenchMAX: A Comprehensive Multilingual Evaluation Suite for Large Language Models.” 2025

In the foundation of modern deep learning models, the Multi Head Attention (MHA) formula plays a crucial role by allowing the model to focus simultaneously on multiple areas of information, thereby improving the ability to capture context within the data sequence (Wei Wang et al., 2024).

$$\begin{aligned}
\text{Attention}(x_i W_{Q_i} = Q_i, x_i W_{K_i} = K_i, x_i W_{V_i} = V_i) &= \\
\text{softmax}\left(\frac{Q_i K_i^T}{\sqrt{M}}\right) V_i & \quad (9) \\
= H_i V_i = H_i [x_i W_{v_i}] &= \hat{H}_i.
\end{aligned}$$

Where:

Q_i (Query) is the query part created from the input segment x_i , which helps determine what should be focused on within the sequence.

K_i (Key) is the key part created from the input segment x_i , used to compare with the query (Q_i) to determine the relevance.

V_i (Value) is the value part, containing the information the model wants to use after calculating the attention score.

$\frac{Q_i K_i^T}{\sqrt{M}}$ is the score calculation between the query and key. This score indicates the level of relevance between parts in the input sequence. Dividing by $\sqrt{M}$ helps reduce the magnitude of the values, making the calculation process more stable.

H_i is the attention weight matrix of head i .

It shows the attention level the model should assign to each part of the input.

$\hat{H}_i = H_i V_i$ is the step of aggregating information from parts of the input according to the attention weight H_i . The result is a new representation of the input, enriched through the attention process.

D. Text Image Input Processing

Mixed-language images pass through a three-agent pipeline—Extract, Analyze, Translate. The first agent OCRs the text, the second interprets it, and the third translates it. Output is a table pairing original and translated text, plus language name and code for quick reference.

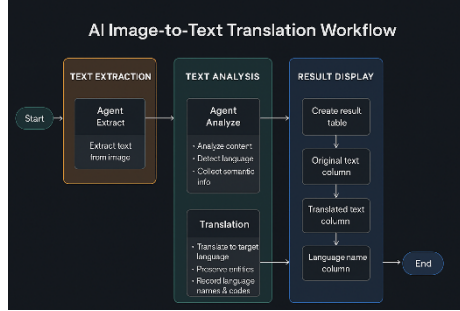


Fig. 8. The Agent process for text image processing.

The study conducted the following experiment to test this AI Agent process.

你好。我叫ana。很高兴认识你。私はベトナムから来ました。 현재 4학년 학생입니다. Le texte que vous lisez est écrit dans de nombreuses langues différentes. souynullonll

Fig. 9. The original image (an image containing multiple mixed languages).

Below are the processing results of the image containing multiple mixed languages.

Language Code	Original Text	Translated Text
zh	你好，我叫	Xin chào, tôi tên là
en	anya.	anya.
zh	，很高兴认识你。	Rất vui được gặp bạn.
ja	私はベトナムから来ました。	Tôi đến từ Việt Nam.
ko	현재 4학년 학생입니다.	Hiện tôi là học sinh lớp 4.
fr	Le texte que vous lisez est écrit dans de nombreuses langues différentes.	Văn bản bạn đang đọc được viết bằng nhiều ngôn ngữ khác nhau.
th	ขอบคุณครับ	Cảm ơn bạn đã đọc.

Fig. 10. The processing results of the image containing multiple mixed languages.

E. Free Form Image Input Processing

Popular OCR tools Tesseract, PaddleOCR, OCRopus, and Google Vision API offer distinct trade offs in accuracy, speed, and customization, so you choose based on project needs. In training, the loss function L (typically Cross Entropy or CTC for OCR) measures the divergence between predicted and target distributions, guiding optimization to boost recognition accuracy.

$$\mathcal{L}_D(f_w) = \frac{1}{|D|} \sum_{(x,y) \in D} L(f_w(x), y) \quad (10)$$

The loss function L outputs a non negative value measuring the gap between the predicted label $\hat{y}$ and the true label y . Larger errors incur larger penalties, so supervised learning aims to minimize this total penalty; when $\hat{y}$ equals y , L drops to 0. The Fourier Transform is a method for transforming a continuous signal from the time domain to the frequency domain.

The formula for the Fourier Transform is as follows:

$$F(w) = \int_{-\infty}^{\infty} p(t) e^{-jwt} dt \quad (11)$$

Where:

$F(w)$ is the result of transforming the signal from the time domain to the frequency domain.

w is the angular frequency, $w=2\pi f$, with the unit of radians per second (rad/s).

j is the imaginary unit (used by electrical and electronics engineers to distinguish it from the symbol for current).

$p(t)$ is the continuous signal in the time domain.

Unlike the Fourier Transform's infinite sine- and cosine-based waves, the wavelet transform uses a finite, well-localized “mother” wavelet. By scaling and translating this mother wavelet, we generate child wavelets—forming the basis of the Continuous Wavelet Transform (CWT).

$$W(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt \quad (12)$$

Where:

$x(t)$: The original signal to be analyzed.

ψ : The mother wavelet function.

- a: The scaling coefficient, adjusting the frequency.
- b: The translation coefficient, adjusting the position in time.
- ψ^* : The complex conjugate of the mother wavelet.

EasyOCR is used here because it supports 80 + languages—including non Latin scripts—installs easily, adapts to varied image conditions, and runs fast without heavy preprocessing (Tison Brokenshire 2025; Renato Tavares 2024).

For single language images, a 7 step, 3 agent pipeline runs: the image is stored; an LLM detects the source language; OCR pinpoints text; the LLM extracts and translates it; the translation is neatly overlaid and merged back into the image; the finished file is saved as result.jpg; and a URL is returned.

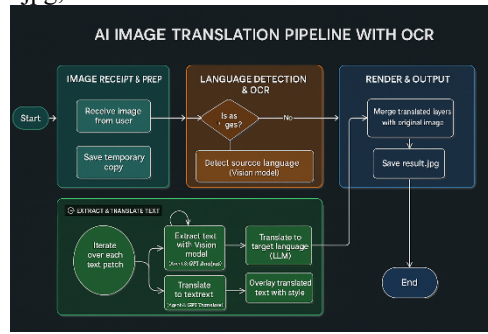


Fig. 11. The Agent process for handling free form text images.

Below are the translation results with the text overlaid directly on the original image.



Fig. 12. The results of free form image processing.

F. Input Format

To implement complex tasks such as file translation and direct image processing in a systematic, transparent, and easily maintainable way, the system is designed based on a pipeline consisting of 15 specialized steps, including 7 AI Agents. Each agent is responsible for a specific task and operates sequentially, forming a closed loop processing and quality control (QA) chain. This approach helps break down the problem, allowing the application of the most suitable AI models for each step and providing the ability to automatically correct errors when quality issues are detected. The diagram below illustrates the entire processing flow of a document file from when it is received to when the final result is returned.

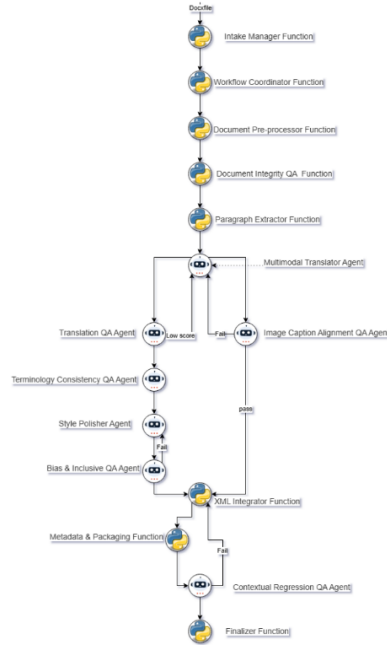


Fig. 13. The image processing model with 15 processing steps and 7 AI agents.

The pipeline begins when Input Management validates the DOCX, records target language/style/temperature, and assigns a translation ID, while Workflow Coordination logs progress and pushes WebSocket updates. Document Pre processing then extracts and normalizes the XML, flags images/tables, and passes it through an Integrity Check (CRC + schema). Verified paragraphs are sent to the Multimodal Translation Agent: GPT 4o handles text, OCR translates any in image text. A Quality Check (BLEU, G BERT) auto retries low scoring segments with a tweaked temperature; Term Check enforces terminology, and Style Refinement (Grok/Gemini Pro) polishes language. Bias and caption accuracy are reviewed by dedicated agents before XML Integration reinserts the refined text and captions. Packaging rebuilds the DOCX, updates metadata, and creates a secure download link, followed by a Regression Check for layout and content integrity. Finally, Completion cleans temporary data, verifies the link, and notifies the user—delivering a self monitoring, high quality translation that preserves formatting and in image text.

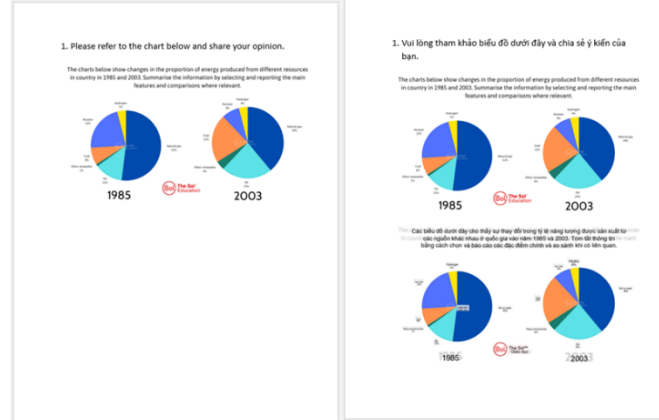


Fig. 14. Direct image and chart processing in the file (with the original file on the left and the result on the right).

4 Experiments

Experiments ran on a Windows laptop (Intel i5 9300H 2.4 GHz, GTX 1650 Ti, 16 GB RAM, 512 GB SSD) using Python 3.11, FastAPI 0.115, and a React (Vite) front end. Key AI libraries included EasyOCR 1.7.2 and LangChain 0.3.20. The test set comprised 32 source pages, each rendered into ~5 translations by five systems: the proposed platform, Google Translate, Microsoft Word, AI TranslateDocs, and OnlineDocTranslator. Six scenarios were evaluated—mixed language pages, format preserving text, narrative, legal, medical, and image rich documents. Three University of Southern Can Tho English experts, blind to system names, scored each output; human judgment was treated as the gold standard.

Results: our system led with an average 8.62/10, followed by AI TranslateDocs 8.30, Microsoft Word 8.08, Google Translate 8.06, and OnlineDocTranslator 7.86. In the mixed language test, it achieved the top single score (8.90), with AI TranslateDocs next at 8.67—demonstrating clear strength in accurately handling multilingual content while maintaining coherence and original formatting.

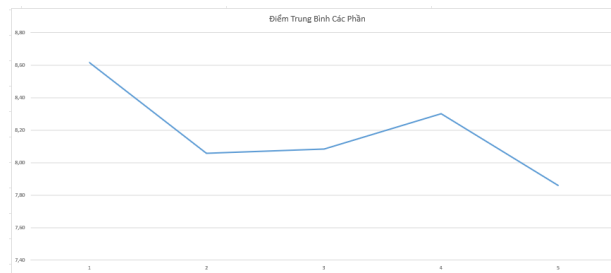


Fig. 15. The average score for all parts of the experiment.

Especially in cases involving images, with text embedded within the image—something that all other programs cannot handle—only the approach proposed in this study successfully processes and translates the embedded text within the image. The following are the results of the text translation embedded in the image.

5 Conclusion

The project met its goal of creating a multifunctional language processing platform that merges existing capabilities with in document image translation. Built on a 15 step pipeline of seven AI agents (e.g., GPT 4o mini, Whisper), the self monitoring architecture boosts accuracy and maintainability. Tests show it outperforms many commercial tools at handling mixed language text, speech, and complex document layouts while preserving images. Next steps include (1) domain specific model tuning and lightweight local deployment, (2) adding real time video translation and more file formats (.pptx, .psd, .xlsx), (3) personalizing style and TTS via voice cloning, and (4) launching a SaaS version with subscriptions, payments, and scalable cloud infrastructure to turn the prototype into a market ready platform.

References

1. Tison Brokenshire (2025), What Makes OCR Technology in 2025 Different? Key Innovations and Top Solutions Compared, accessed on 5/6/2025.
2. Xu Huang et al. (2025). BenchMAX: A Comprehensive Multilingual Evaluation Suite for Large Language Models. arXiv preprint arXiv:2502.07346.
3. Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., ... & Sifre, L. (2022). Training compute optimal large language models. arXiv preprint arXiv:2203.15556.
4. Melvin Johnson et al. (2017). Google’s multilingual neural machine translation system: Enabling zero shot translation. *Transactions of the Association for Computational Linguistics*, 5, 339–351.
5. Rebecca Knowles et al (2024). Calibration and context in human evaluation of machine translation. *Natural Language Processing*, 1–25.
6. Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling laws for neural language models. arXiv preprint arXiv:2001.08361.
7. Nguyễn Thị Việt Nga (2022), Công nghệ dịch thuật và vai trò của biên dịch viên trong cuộc Cách mạng công nghiệp 4.0. *Tạp chí Lý luận chính trị và Truyền Thông*, số 03/2022.
8. Thi Tuyet Hai Nguyen et al. (2021). Survey of post OCR processing approaches. *ACM Computing Surveys (CSUR)*, 54(6), 1–37.
9. Zhibin Lan et al. (2023). Exploring better text image translation with multimodal codebook. arXiv preprint arXiv:2305.17415.
10. Michael McGuire (2025). Automatic speech recognition for non native English: Accuracy and disfluency handling. arXiv preprint arXiv:2503.06924.
11. OpenAI (2022), Introducing Whisper. <https://openai.com/index/whisper/>, accessed on 5/6/2025.
12. Nick Ovchinnikov (2024), Machine Learning Text To Speech: Intro, Little Theory and Math, accessed on 5/6/2025.

13. Xudong Pan et al. (2018, July). Theoretical analysis of image to image translation with adversarial learning. In International Conference on Machine Learning (pp. 4006 4015). PMLR.
14. Miguel Angel Rios Gaona et al. (2023, June). Quality Analysis of Multilingual Neural Machine Translation Systems and Reference Test Translations for the English Romanian language pair in the Medical Domain. In Proceedings of the 24th Annual Conference of the European Association for Machine Translation (pp. 355 364).
15. Giuseppe Sofo (2024). Reshaping Translation and Theories of Translation in the Digital Age. *Status Quaestionis*, (26).
16. Ilya Sutskever et al. (2014). Sequence to sequence learning with neural networks. *Advances in neural information processing systems*, 27.
17. Renato Augusto Tavares (2024). Comparison of Image Preprocessing Techniques for Vehicle License Plate Recognition Using OCR: Performance and Accuracy Evaluation. arXiv preprint arXiv:2410.13622.
18. Ayushi Trivedi et al. (2018). Speech to text and text to speech recognition systems Areview. *IOSR J. Comput. Eng.*, 20(2), 36 43.
19. Wei Wang et al (2024). Universal Approximation Theory: The Basic Theory for Transformer based Large Language Models. arXiv preprint arXiv:2407.00958.
20. ViettelAI (2025), Lịch Sử Và Xu Hướng Tương Lai Của Công Nghệ Chuyển Văn Bản Thành Giọng Nói. <https://viettelai.vn/tin-tuc/lich-su-phat-trien-cong-nghe-chuyen-doi-van-ban-thanh-giong-noi>, ngày truy cập 5/6/2025.
21. Yonghui Wu et al. (2016). Google's neural machine translation system: Bridging the gap between human and machine translation. arXiv preprint arXiv:1609.08144.
22. Wayne Xin Zhao et al. (2023). A survey of large language models. arXiv preprint arXiv:2303.18223, 1(2).